

Product Requirements Document

AI-Powered Payment Failure Triage Assistant

Author: Aastha Devadiya

Version: 1.5

Date: June 2026

Status: Draft

1. Problem Statement

When a card or net banking transaction fails, customers are often left without clarity. They see a generic error message, are unsure whether the amount was debited, and receive no guidance on the next steps. This leads to:

- Repeated transaction attempts, often worsening the issue
- Increased call centre volume for issues that could be self-resolved
- Customer frustration and reduced trust in digital channels

Today, card networks (Visa, Mastercard, American Express, RuPay) and fintech issuers rely on generic decline codes that provide little value to end users. There is no intelligent, customer-facing layer that interprets the reason for a payment failure and delivers clear, actionable guidance in plain language, instantly.

2. Objective

Build an intelligent, LLM-powered web application that:

1. Accepts a payment failure scenario through structured inputs (dropdown) and optional free-text details.
2. Identifies the most probable root cause, including network- or bank-specific behaviour where applicable.
3. Provides a clear, actionable resolution in simple, customer-friendly language.
4. Demonstrates how card networks and fintech companies can reduce customer support dependency through intelligent self-service.

3. Scope

In Scope

- Payment modes: Payments (A2A and Bill Payments) and Cards
- Platforms covered:
 - Mobile Banking App (bank's own app)
 - Web Banking (browser-based net banking)
 - Third-party apps using saved cards (Google Pay, PhonePe, Paytm card-on-file)
- Platform-specific factors:
 - Authentication method (biometric / MPIN on app vs OTP / password / QR on web)
 - Transaction limits (limits often differ between mobile app and web banking)
- Card networks covered: Visa, Mastercard, American Express, RuPay
- Banks covered:
 - Private Banks: HDFC, ICICI, Axis, Kotak, Yes Bank
 - Public Sector Banks: SBI, PNB, Bank of Baroda, Canara Bank, Union Bank
 - Small Finance Banks: AU Small Finance, Equitas, ESAF, Jana, Ujjivan
 - Other: free text input for unlisted banks

- Failure types covered:
 - Authentication failures (wrong credentials, device binding, QR generation/expiry, MPIN, OTP not received)
 - Transaction limit breaches (daily/monthly limits exceeded)
 - Technical / system timeouts (client - server errors, gateway failures)
 - Card blocked or expired
 - Incomplete personal details
 - Insufficient funds / credit limit exceeded

Out of Scope (V1)

- UPI payments
- International transactions and forex failures
- Chargeback or dispute flows
- Integration with any live banking system or real account data
- Wallet-based payments (Paytm, PhonePe etc.)

4. User Personas

Persona 1 — Riya, 28, Salaried Professional

Uses net banking and her HDFC Visa debit card for rent and bill payments. Not very tech-savvy. Gets anxious when a payment fails, especially if she is unsure whether money was deducted. Wants a clear, reassuring answer - not technical jargon.

Persona 2 — Arjun, 35, Small Business Owner

Makes 15-20 Mastercard credit card transactions per week for vendor payments. When a transaction fails mid-flow, he needs to know immediately whether to retry or use another method. Values speed over explanation.

Persona 3 — Priya, 22, First-Time Amex User

Just received her first American Express card. Unfamiliar with Amex-specific acceptance limitations and authentication flows. Needs the tool to educate, not just resolve.

5. User Flows

The tool adapts its flow based on which type of organization deploys it. Each segment has a different starting context, different information already available, and different user needs.

Flow A - Bank Deployment

(e.g. ICICI Bank, HDFC Bank, SBI embed this in their own app or web banking)

Context already known: Bank name, customer's card network, payment mode **What the bot skips:** Bank selection (already known from login context)

Tone: Specific, reassuring, action-oriented

Step 1: User lands on the tool via their bank's app or web banking

↓

Step 2: Selects failure type from dropdown

(Authentication / Limit Breach / Timeout / Card Issue / Insufficient Fund / Incomplete Personal Details)

↓

Step 3: Free text box appears – "Tell us what happened"

(e.g. "My card was declined at checkout but I have sufficient balance")

↓

Step 4: LLM processes platform + failure type + free text

(Bank and card network pre-filled from deployment context)

↓

Step 5: Tool returns:

- Likely cause specific to that bank's known behavior
- Immediate action steps using that bank's specific app/portal navigation
- Whether to retry or wait
- Direct link or number to that bank's support team

Why this flow is shorter: The bank already knows who the customer is and what card they hold. Asking them to re-enter this is friction - and bad UX.

Note on platform context: Platform context (mobile app vs web banking) is passed automatically via the bank's API integration at deployment — the user does not need to select it manually.

Flow B — Card Network Deployment

(e.g. Visa, Mastercard, American Express, RuPay embed this in their cardholder support portal)

Context already known: Card network

What the bot focuses on: Network-layer failures — decline codes, 3DS issues, acceptance problems

Tone: Technical but clear, focused on network vs issuer distinction

Step 1: User lands on the tool via the network's support page

↓

Step 2: Selects their issuing bank from dropdown
 (Private: HDFC, ICICI, Axis, Kotak, Yes Bank
 Public: SBI, PNB, Bank of Baroda, Canara, Union Bank
 Small Finance: AU, Equitas, ESAF, Jana, Ujjivan
 + "other" with free text)

↓

Step 3: Selects platform where failure occurred
 (Mobile Banking App / Web Banking / Google Pay / PhonePe / other third-party app)

↓

Step 4: Selects payment mode (Debit Card / Credit Card)

↓

Step 5: Selects failure type from dropdown
 (Authentication / Limit Breach / Timeout / Card Issue / Insufficient Fund)

↓

Step 6: Free text box appears – "Describe what happened"

↓

Step 7: LLM processes bank + platform + payment mode + failure type + free text
 (Card network pre-filled from deployment context)

↓

Step 8: Tool returns:

- Clear distinction: is this a network-level issue or an issuing bank
- If network issue: step-by-step resolution
- If bank issue: redirect to issuing bank with clear explanation of why
- Acceptance-specific guidance where relevant (e.g. Amex merchant)

Why the network/bank distinction matters: The biggest frustration for cardholders is being bounced between Visa and their bank with no clarity on who owns the problem. This flow resolves that explicitly.

Flow C — Third-Party App Deployment

(e.g. Google Pay, PhonePe, Paytm embed this in their help or support section)

Context already known: Platform (the third-party app itself)

What the bot focuses on: Full context needed - these apps sit furthest from the transaction

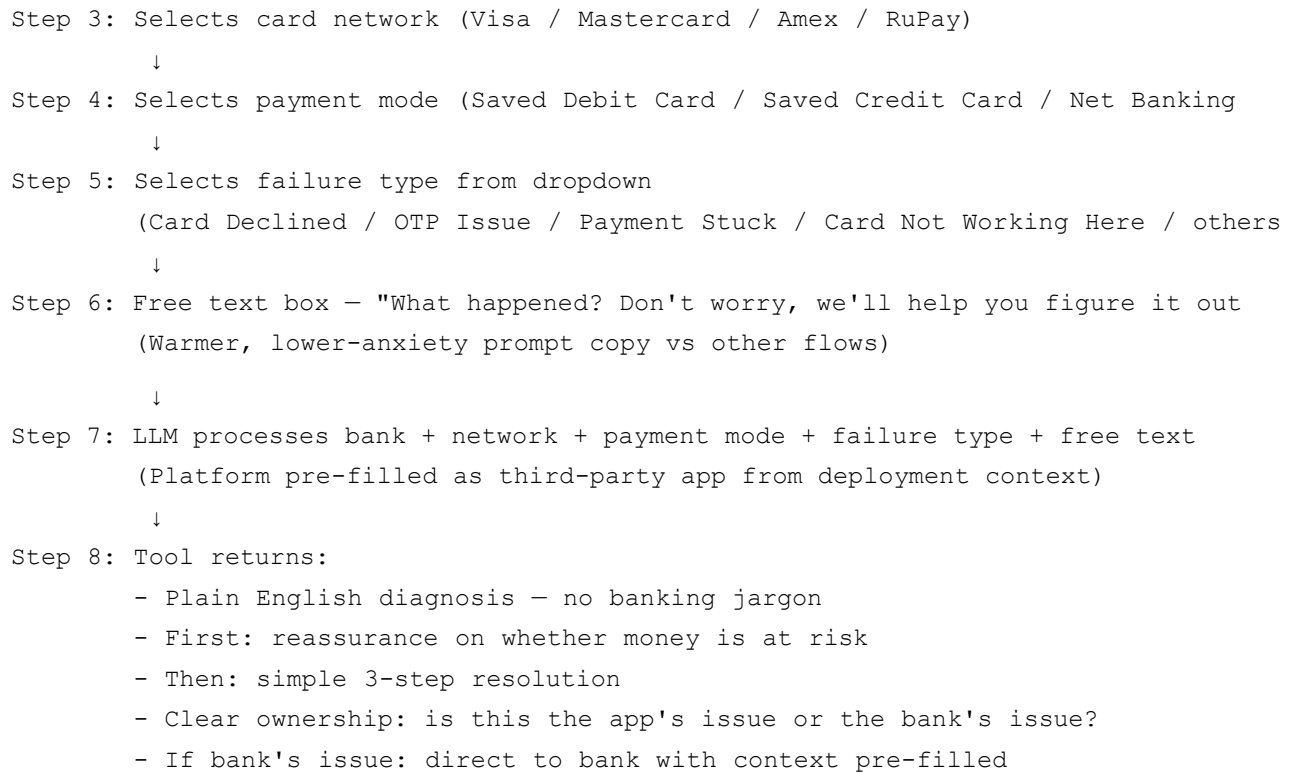
Tone: Simplest language, most reassuring, assumes least technical knowledge

Step 1: User lands on the tool via the third-party app's help section

↓

Step 2: Selects their bank from dropdown
 (Private: HDFC, ICICI, Axis, Kotak, Yes Bank
 Public: SBI, PNB, Bank of Baroda, Canara, Union Bank
 Small Finance: AU, Equitas, ESAF, Jana, Ujjivan
 + "other" with free text)

↓



Why tone differs here: GPay and PhonePe users skew younger and less familiar with banking infrastructure. Anxiety about stuck money is higher. The bot needs to reassure before it resolves.

C. Failure Scenarios & Expected Responses

Core Scenarios (All Networks)

Failure Type	Example User Input	Expected Bot Behaviour
Authentication - OTP not received	"I didn't get the OTP on my registered number"	Diagnose: SIM/network issue or bank delay. Steps: wait 2 mins, resend OTP, check DND status, try alternate method
Authentication - Wrong PIN	"I entered my PIN but it said incorrect"	Warn: 3 wrong attempts = card block. Steps: verify PIN, use forgot PIN flow before retrying
Transaction limit breach	"Payment failed but I have enough balance"	Diagnose: daily/monthly limit hit. Steps: check limit in app, split payment, or request limit increase
System timeout	"Page froze and I don't know if money was deducted"	Reassure first: explain pending state. Steps: check statement, wait 10-15 mins, do not retry immediately
Card blocked	"My card is declining everywhere"	Diagnose: block due to suspicious activity or expiry. Steps: check expiry date, call bank helpline, request replacement

Insufficient funds	“Transaction failed even though I thought I had balance”	Diagnose: available balance vs. ledger balance difference. Steps: check hold amounts, check statement
Incomplete Personal Details	“All of my personal details are updated but showing incomplete details error”	Diagnose: Complete personal info. including KYC and Demographic details. Steps: Check field name, suggest way to update.

Network-Specific Scenarios

Network	Unique Failure Type	Why It's Different	Expected Bot Behavior
American Express	Low merchant acceptance	Amex operates a closed-loop network - fewer merchants accept it vs Visa/Mastercard	Explain closed-loop model. Suggest switching to Visa/Mastercard for that merchant. Reassure card is not blocked
American Express	Higher decline rate on international sites	Amex has stricter fraud thresholds on cross-border transactions	Advise enabling international usage via Amex app; explain it's a security feature not a card issue
RuPay	Failure on international platforms	RuPay has limited global acceptance outside India and select partner countries	Explain RuPay's domestic focus. Advise using Visa/Mastercard for international transactions
Mastercard	3D Secure authentication loop	Mastercard's 3DS2 flow can loop on certain browsers/devices	Steps: clear cache, try incognito mode, switch browser, disable VPN
Visa	Verified by Visa OTP mismatch	Visa's OTP is issued by issuing bank, not Visa - customers often blame Visa	Clarify that OTP comes from their bank, not Visa. Direct to bank helpline
RuPay	NFC / contactless failure on POS terminal	RuPay's NFC and contactless behaviour differs from Visa/Mastercard on certain POS terminals — some terminals do not yet	Diagnose terminal compatibility issue. Advise user to try chip+PIN as fallback. Clarify that the card is not blocked — it is a terminal-level acceptance gap.

7. LLM Prompt Design (Core Logic)

The tool will send the following context to the LLM:

```
System prompt:
"You are a helpful payment support assistant with deep knowledge of card networks including Visa, Mastercard, American Express, and RuPay. A customer has experienced a payment failure. Based on the issuing bank, platform used (mobile app, web banking, or third-party app), card network, payment mode, failure type, and their description, diagnose the most likely cause and provide a clear, step-by-step resolution in simple, non-technical language.

Where relevant, factor in network-specific behaviour (e.g. Amex acceptance limitations, RuPay international restrictions, Mastercard 3DS flows).

Always reassure the customer first if money may be at risk. Never ask for or acknowledge sensitive information like account numbers, card numbers, or password. Add a disclaimer that this is general guidance and account-specific queries should be directed to their bank. keep the response under 150 words."

User message:
"Bank: [HDFC Bank]
Platform: [Mobile Banking App]
Payment mode: [Credit Card]
Card network: [American Express]
Failure type: [Card declined at merchant]
Customer description: [My Amex card was declined at a restaurant but my Visa card worked fine at the same place]"
```

Platform-Specific Scenarios

Platform	Failure Type	Why It's Different	Expected Bot Behaviour
Mobile App	Biometric authentication failure	App uses fingerprint/face ID instead of OTP - failure reasons differ entirely from web	Advise resetting biometric in app settings; fallback to MPIN; clarify this is device-level not bank-level issue
Mobile App	Transaction limit lower than expected	Many banks set lower default limits on mobile app vs web banking for security	Inform user that mobile app limits are often lower by design; direct to web banking for higher-value transactions or to request limit increase

Web	OTP not received on	Web banking relies on SMS OTP; mobile app uses in-app or	Diagnose SMS delivery issue; advise checking spam, DND status, network signal; suggest
Banking	browser	biometric auth - different delivery path	switching to mobile app as alternate
Web Banking	Session timeout mid-transaction	Browser sessions expire faster than app sessions; payment may be in pending state	Reassure on money status; advise checking statement before retrying; suggest mobile app for future transactions
Google Pay / PhonePe	Saved card declined	Third-party apps use tokenised card credentials - failure may be at token layer, not card layer	Explain tokenization; advise re-adding card in the app; clarify that the card itself is not blocked
Google Pay / PhonePe	Bank authentication page not loading	Third-party app redirects to bank's 3DS page — failure is on bank's end, not the app	Direct user to bank's own app or web banking to verify; clarify GPay/PhonePe are not responsible for this failure

Bank-Specific Scenarios

Bank Type	Unique Behaviour	Expected Bot Behaviour
SBI	Net banking OTP via SMS can be delayed due to high volume	Advise waiting longer than usual; suggest YONO app as alternate channel
HDFC	OTP sometimes routed via third-party SMS aggregator causing delay	Suggest checking spam folder; advise using HDFC app authenticator instead
Axis	Daily debit card limit lower than other private banks by default	Inform user of default limit; direct to Axis app to increase limit instantly
AU Small Finance	Limited merchant terminal acceptance vs large private banks	Explain small finance bank acceptance gaps; suggest carrying backup card
Other / Unlisted	Unknown bank-specific behaviour	Provide general resolution steps; advise contacting their bank's helpline directly

6. Success Metrics

Metric	Target (V1)
Scenario coverage	6 core + 5 network-specific scenarios handled accurately
Response accuracy	90%+ correct diagnosis when tested manually across all scenarios
Network awareness	Bot correctly identifies network-specific nuance in 100% of network-specific test cases
Bank awareness	Bot references bank-specific behaviour for at least 5 major banks accurately
Platform awareness	Bot correctly differentiates mobile app vs web banking vs third-party app failures in 100% of platform-specific test cases
Response clarity	Understandable to a non-technical user (validated via 3 user tests)
Live deployment	Accessible via public Streamlit URL
Response time	Under 5 seconds per query

7. Revenue Impact

Why This Tool Has Direct Business Value

This tool impacts both revenue generation and cost reduction - making it relevant to product, business, and finance stakeholders at any bank, card network, or fintech.

- **Revenue Generation**

Failed transactions = lost revenue at every layer of the payment stack

Every failed transaction that goes unresolved represents lost interchange, lost GMV, and a customer who may abandon the merchant entirely.

Stakeholder	Revenue Impact of Unresolved Failures
Card Networks (Visa, Mastercard, Amex)	Lost interchange fee on every failed transaction that is not retried
Issuing Banks	Lost transaction fee revenue; potential customer churn if failures are frequent
Third-Party Apps (GPay, PhonePe)	Lost MDR (Merchant Discount Rate) on failed transactions; lower user retention
Merchants	Lost sale if customer abandons checkout after failure

How this tool recovers revenue:

- Guides users to retry correctly instead of abandoning - directly recovers failed transaction value
- Faster resolution = higher retry success rate within the same session
- Reduces customer churn caused by repeated unexplained failures

Directional model (illustrative):

- Assume 1% of monthly transactions on a platform fail = ~550,000 failures/month for a 55M user base
- If the bot successfully resolves and enables retry on 20% of those = 110,000 recovered transactions/month
- At an average transaction value of ₹2,000 = ₹22 Cr+ in recovered GMV monthly

- **Cost Reduction**

Call centre deflection is the clearest, most measurable saving

Metric	Assumption
Average cost per call centre interaction	₹150 – ₹300
% of call centre volume from payment failures	~25–35% (industry estimate)
Bot self-resolution rate (target)	30% of payment failure queries
Users on platform	55M

Outcome: Even a 30% deflection rate on payment failure calls translates to significant cost saving at scale — and the bot also reduces average handle time for escalated calls because it pre-fills failure context before handoff to an agent.

Additional cost benefits:

- Reduces repeat contacts — customers who understand why a failure happened are less likely to call again
- Lowers agent training burden — complex network vs bank ownership questions are handled by the bot, not the agent
- Strategic Value for Each Deployment Segment

Segment	Primary Value
Banks	Call centre cost deflection + customer retention
Card Networks	Interchange recovery + cardholder satisfaction scores
Third-Party Apps	Retry conversion + reduced churn on payment failures

1. Competitive Differentiation

How This Is Different from Existing Chatbots

Most banks, networks, and fintech apps already have chatbots. The honest question is: why does this tool need to exist?

The answer: existing chatbots solve for containment, not resolution.

They are designed to reduce call volume by deflecting queries — not to actually diagnose and fix the problem. The result is frustrated customers who feel the bot is a barrier to reaching a human, not a tool that helps them

- Head-to-Head Comparison

Capability	Existing Bank / Fintech Chatbots	This Tool
Input	Keyword matching or rigid	Free text + structured dropdowns —
method	menu trees	understands natural language
Diagnosis depth	Generic: “Your transaction failed. Please try again.”	Specific: explains <i>why</i> it failed based on bank + platform + network + failure type
Ownership clarity	Never clarifies if failure is bank’s or network’s fault	Explicitly tells the user who owns the problem - bank, network, or app (applies specifically to third-party app deployment context)
Platform awareness	Same response regardless of mobile app or web banking	Different diagnosis and resolution steps based on platform
Network awareness	No differentiation between Visa, Mastercard, Amex, RuPay	Network-specific guidance - Amex acceptance, RuPay international limits, Mastercard 3DS
Deployment adaptability	One fixed flow for all users	Three separate flows - bank, network, third-party - each optimized for context
Escalation quality	Dumps user to agent with no context	Pre-fills failure context before handoff - agent knows the issue before picking up
Tone adaptability	Same tone across all users	Tone adapts by segment - technical for network users, reassuring for third-party app users

- What This Tool Is Not Trying To Replace

This tool is not a full customer support platform. It is a first-line intelligent triage layer that:

- Resolves what can be self-served

- Pre-qualifies what cannot
- Hands off with full context to existing support systems

This positioning means it complements existing chatbots and CRM systems - it does not compete with them. That makes adoption by banks and networks significantly easier.

2. Technical Stack

Component	Tool	Why
Frontend + deployment	Streamlit	Free, Python-based, no frontend coding needed
LLM	Claude API (claude-sonnet-4-6) or OpenAI GPT-4o	Best reasoning for nuanced, network-aware diagnosis
Code generation	Cursor AI	Writes and fixes all code based on PRD prompts — no coding required
Version control	GitHub	Proof of iterative builds; portfolio credibility
Screen recording	Loom	Demo video for LinkedIn and interviews
Hosting	Streamlit Community Cloud	Free public URL, one-click deploy from GitHub

3. Future Considerations (V2 Backlog)

- Multilingual support (Hindi, regional languages)
- Voice input for accessibility
- Coverage of UPI, wallets, and international transactions
- Analytics dashboard showing most common failure types by network
- White-label version demonstrating how Visa, Mastercard, or Amex could embed this in their customer support flow
- WhatsApp bot version

4. Risks & Mitigations

Risk	Mitigation
LLM gives incorrect advice (e.g. says money not deducted when it was)	Add disclaimer: "This tool provides general guidance only. Please verify with your bank statement or contact your bank/card network for account-specific queries."
User inputs sensitive data (card number, account number, password)	System prompt instructs LLM to never acknowledge or use such data; add UI warning before free text box
Network-specific advice becomes outdated (e.g. RuPay expands internationally)	Add "last updated" date on tool; treat network behaviour as a living knowledge layer
Scope creep during build	Strictly follow V1 scope; log all new ideas as V2 backlog

PRD prepared by Aastha Devadiya as part of personal AI product portfolio.

This is an independent proof-of-concept project. No proprietary or internal bank data has been used

